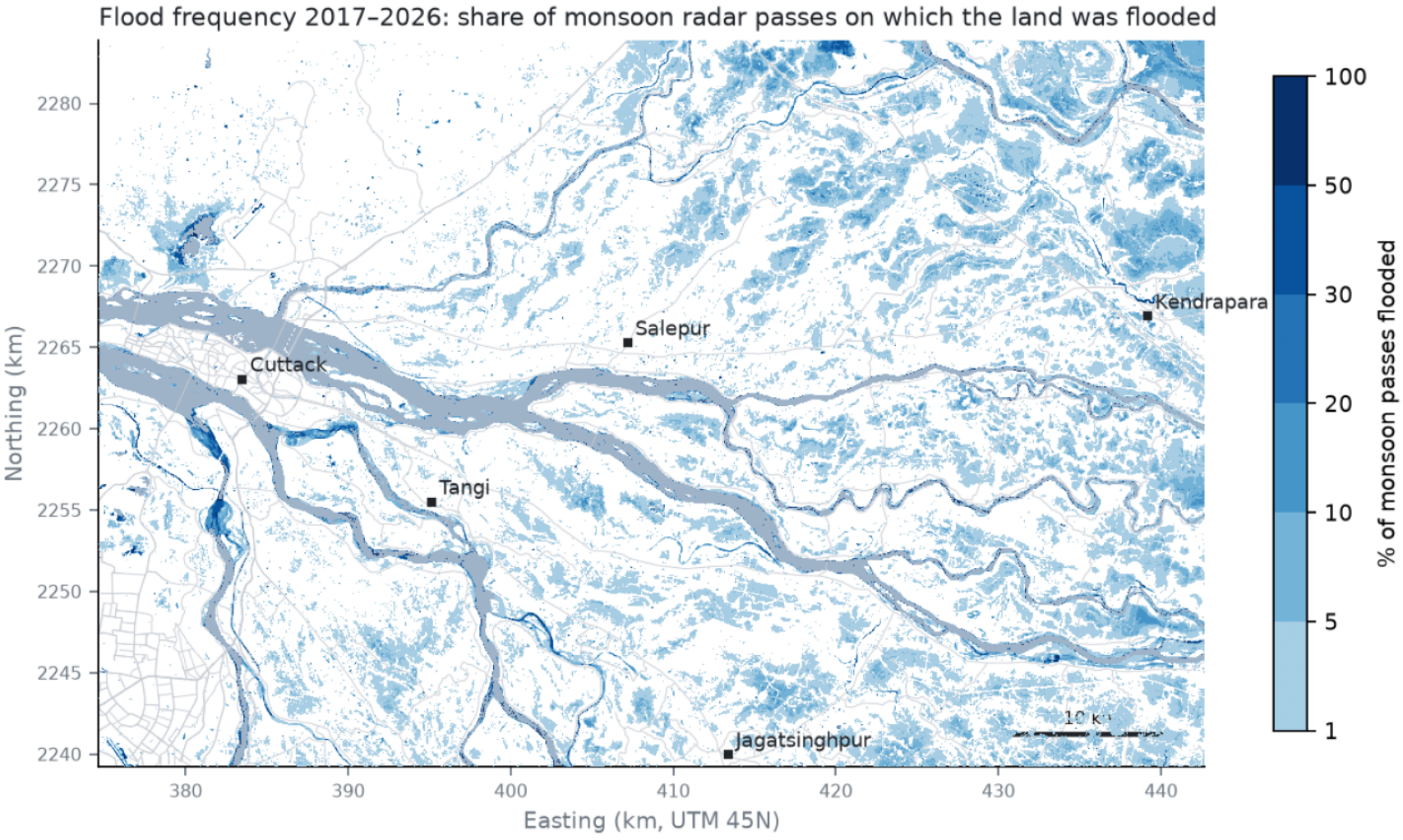
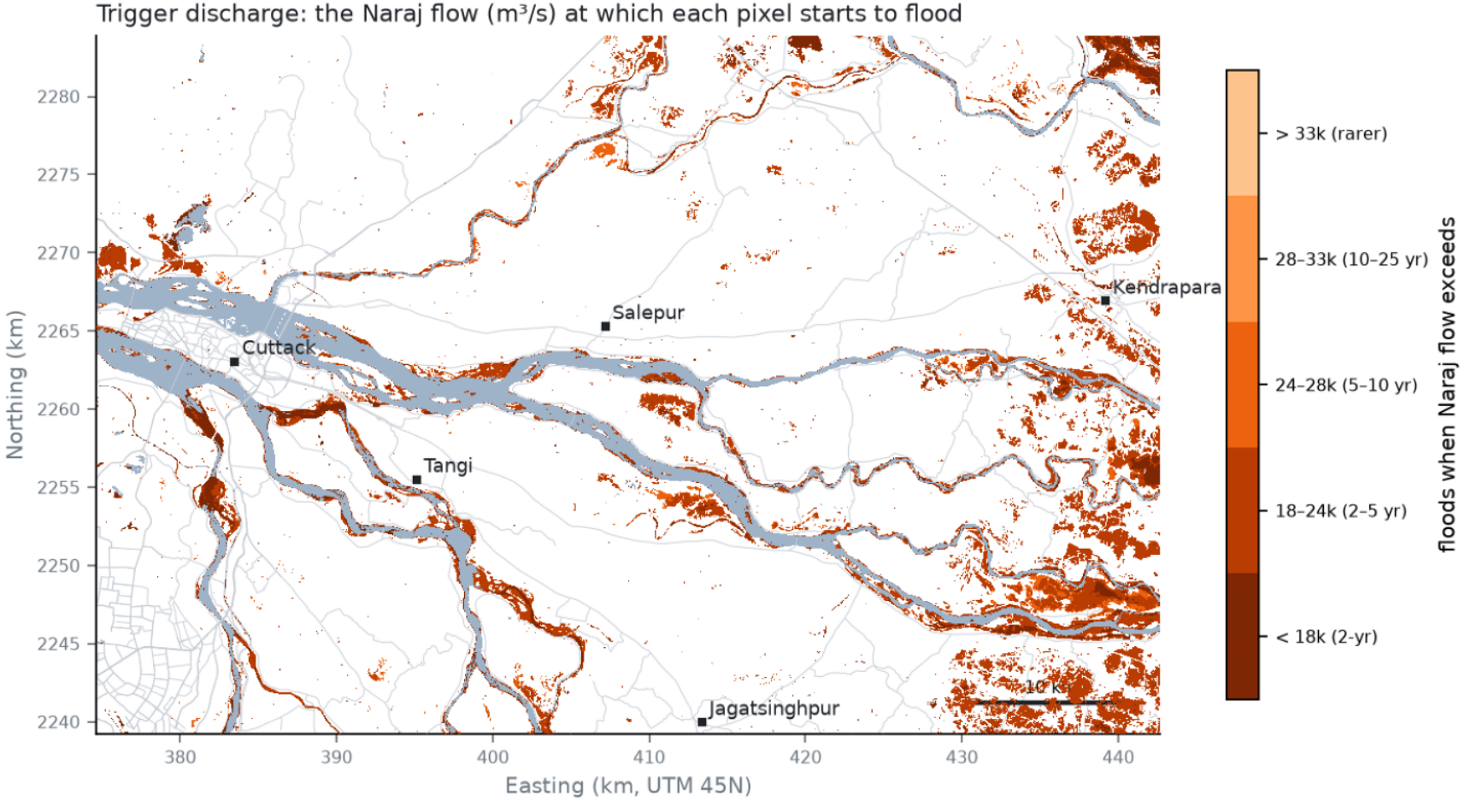
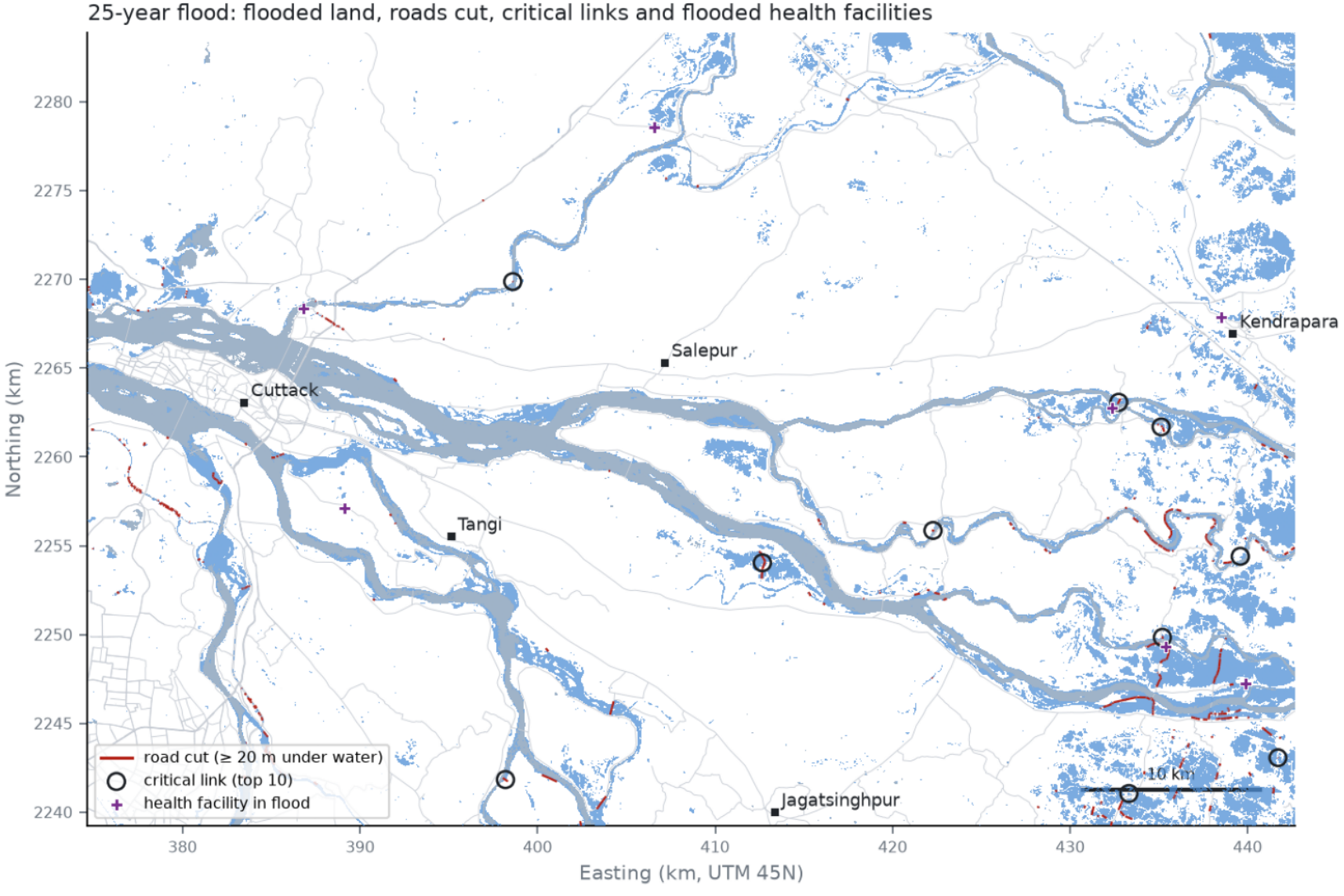
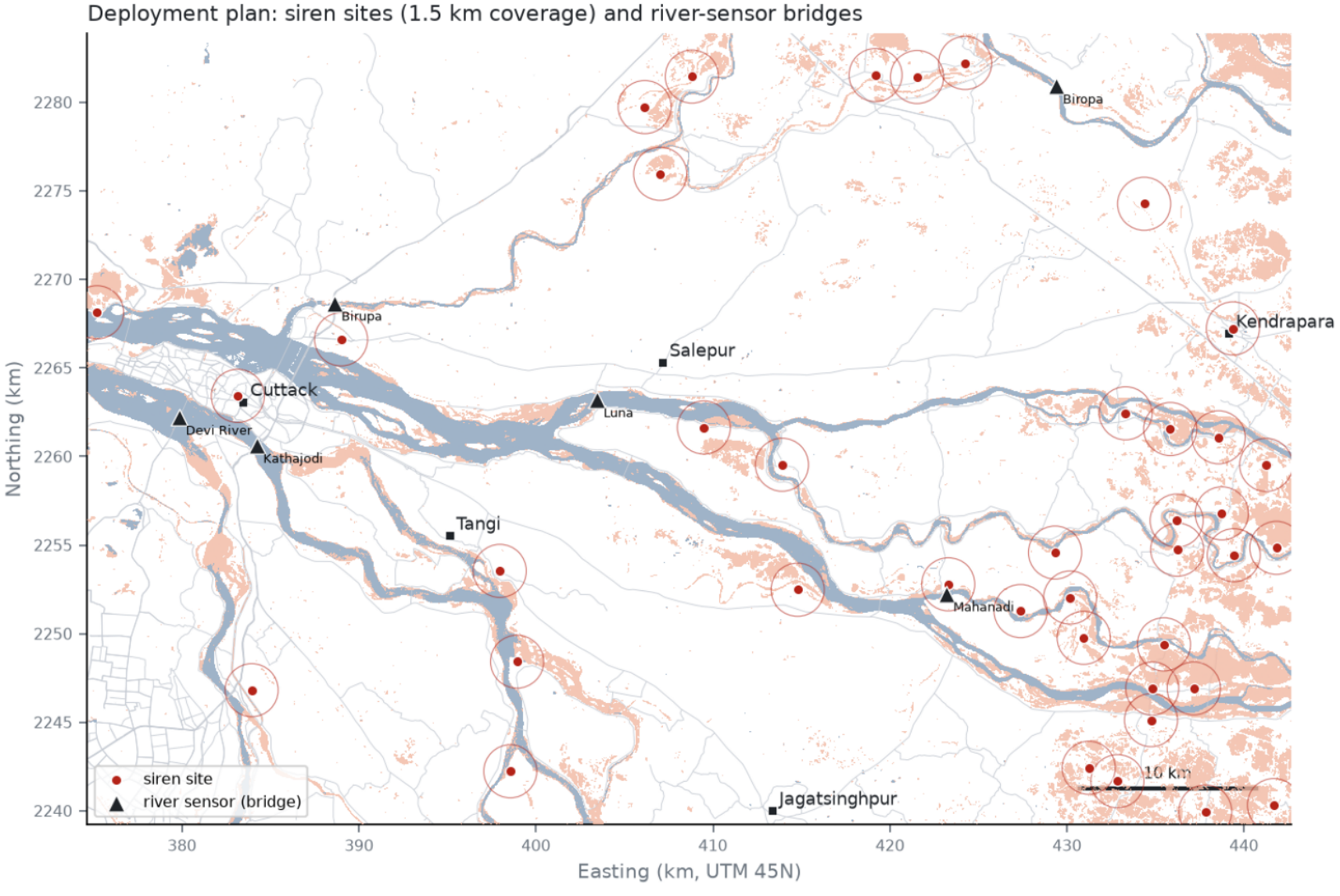
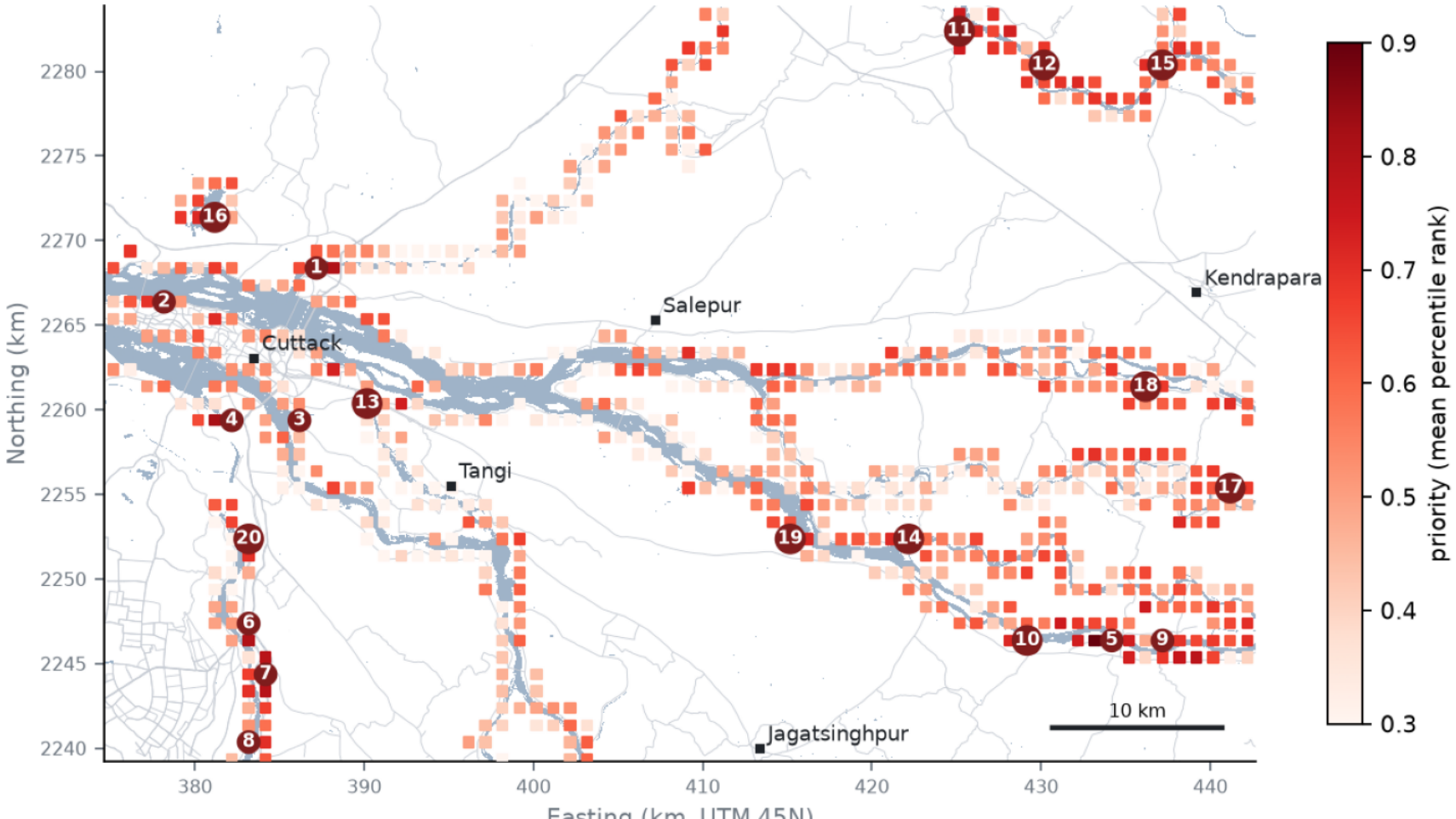

Flood frequency 2017–2026: share of monsoon radar passes on which the land was flooded
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
% of monsoon passes flooded
100
50
30
20
10
5
1


Trigger discharge: the Naraj flow (m³/s) at which each pixel starts to flood
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
> 33k (rarer)
28–33k (10–25 yr)
24–28k (5–10 yr)
18–24k (2–5 yr)
< 18k (2-yr)
floods when Naraj flow exceeds


25-year flood: flooded land, roads cut, critical links and flooded health facilities
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
road cut (≥ 20 m under water)
critical link (top 10)
health facility in flood
10 km
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440


Deployment plan: siren sites (1.5 km coverage) and river-sensor bridges
Northing (km)
Easting (km, UTM 45N)
2280
2275
2270
2265
2260
2255
2250
2245
2240
380
390
400
410
420
430
440
Biropa
Birupa
Kendrapara
Salepur
Cuttack
Luna
Devi River
Kathajodi
Tangi
Mahanadi
Jagatsinghpur
siren site
river sensor (bridge)
10 km


River-bank inspection priority (1 km cells; field list = top 20 at least 3 km apart)
Northing (km)
Easting (km, UTM 45N)
priority (mean percentile rank)
Cuttack
Salepur
Kendrapara
Tangi
Jagatsinghpur
10 km

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GT01	ground truth	20.53991	86.029907	atlas: never flooded; ask in near Tangi (4.1 km)
GT02	ground truth	20.329164	86.377417	atlas: never flooded; ask in near Patkura (7.3 km)
GT03	ground truth	20.525954	86.088884	atlas: never flooded; ask in near Salepur (5.1 km)
GT04	ground truth	20.448769	85.869244	atlas: never flooded; ask in near Badambadi (2.7 km)
GT05	ground truth	20.411891	86.365583	atlas: never flooded; ask in near Patkura (4.7 km)
GT06	ground truth	20.304466	85.939058	atlas: never flooded; ask in near Bhingarapur (4.2 km)
GT07	ground truth	20.387887	86.053668	atlas: never flooded; ask in near Kishannagar (4.7 km)
GT08	ground truth	20.25097	85.93789	atlas: never flooded; ask in near Bhingarapur (1.6 km)
GT09	ground truth	20.318284	85.894331	atlas: never flooded; ask in near Balikanthia (4.3 km)
GT10	ground truth	20.356498	86.170934	atlas: never flooded; ask in near Nemalo (8.2 km)
GT11	ground truth	20.279583	86.311921	atlas: rarely (<10 %); ask in near Jagatsinghpur (14.5 km)
GT12	ground truth	20.441777	86.438126	atlas: rarely (<10 %); ask in near Patkura (5.8 km)
GT13	ground truth	20.39213	86.399014	atlas: rarely (<10 %); ask in Patkura
GT14	ground truth	20.463825	85.996448	atlas: rarely (<10 %); ask in near Jagatpur (6.3 km)
GT15	ground truth	20.288905	86.292918	atlas: rarely (<10 %); ask in near Jagatsinghpur (12.6 km)
GT16	ground truth	20.621271	86.347445	atlas: rarely (<10 %); ask in near Nikirai (6.5 km)
GT17	ground truth	20.633745	86.258896	atlas: rarely (<10 %); ask in near Balachandrapur (2.7 km)
GT18	ground truth	20.39787	86.337273	atlas: rarely (<10 %); ask in near Patkura (7.6 km)
GT19	ground truth	20.423643	86.419021	atlas: rarely (<10 %); ask in near Patkura (4.5 km)
GT20	ground truth	20.55264	86.358097	atlas: rarely (<10 %); ask in near Dhuria (2.8 km)
GT21	ground truth	20.345835	85.879191	atlas: sometimes (10-30 %); ask in near Balikanthia (7.9 km)
GT22	ground truth	20.402776	85.970394	atlas: sometimes (10-30 %); ask in near Kandarpur (4.7 km)
GT23	ground truth	20.288954	86.037798	atlas: sometimes (10-30 %); ask in near Gobindpur (4.1 km)
GT24	ground truth	20.573884	86.132349	atlas: sometimes (10-30 %); ask in near Mahanga (3.9 km)
GT25	ground truth	20.500326	85.935606	atlas: sometimes (10-30 %); ask in Jagatpur
GT26	ground truth	20.463742	86.150418	atlas: sometimes (10-30 %); ask in near Salepur (4.5 km)
GT27	ground truth	20.64158	86.19282	atlas: sometimes (10-30 %); ask in near Balachandrapur (5.5 km)
GT28	ground truth	20.630124	86.140477	atlas: sometimes (10-30 %); ask in near Mahanga (7.5 km)
GT29	ground truth	20.52742	86.37988	atlas: sometimes (10-30 %); ask in Dhuria
GT30	ground truth	20.637493	86.290745	atlas: sometimes (10-30 %); ask in near Balachandrapur (6.3 km)
GT31	ground truth	20.27676	85.900371	atlas: often (>=30 %); ask in Balikanthia
GT32	ground truth	20.444484	86.383472	atlas: often (>=30 %); ask in near Fakirabad (6.7 km)

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GT33	ground truth	20.43527	86.167062	atlas: often (>=30 %); ask in Nemalo
GT34	ground truth	20.508202	86.414099	atlas: often (>=30 %); ask in Fakirabad
GT35	ground truth	20.360662	86.296041	atlas: often (>=30 %); ask in near Patkura (12.1 km)
GT36	ground truth	20.249727	85.883137	atlas: often (>=30 %); ask in near Balikanthia (3.6 km)
GT37	ground truth	20.427815	86.085821	atlas: often (>=30 %); ask in Kishannagar
GT38	ground truth	20.262807	86.015363	atlas: often (>=30 %); ask in near Gobindpur (7.2 km)
GT39	ground truth	20.441254	86.064461	atlas: often (>=30 %); ask in near Kishannagar (3.8 km)
GT40	ground truth	20.547107	85.820947	atlas: often (>=30 %); ask in near Gurudijhatia (2.4 km)
SR01	siren site	20.50289	86.41865	rank 1: Fakirabad, covers 1,794 more people
SR02	siren site	20.31901	86.37564	rank 2: near Patkura (8.5 km), covers 1,160 more people
SR03	siren site	20.33115	86.03182	rank 3: near Gobindpur (1.1 km), covers 787 more people
SR04	siren site	20.3872	86.41953	rank 4: near Patkura (1.5 km), covers 760 more people
SR05	siren site	20.63012	86.12477	rank 5: near Bhattasagar Square (7.0 km), covers 669 more people
SR06	siren site	20.63049	86.24689	rank 6: near Balachandrapur (1.4 km), covers 555 more people
SR07	siren site	20.40471	86.38856	rank 7: near Patkura (2.4 km), covers 554 more people
SR08	siren site	20.63768	86.27285	rank 8: near Balachandrapur (3.8 km), covers 505 more people
SR09	siren site	20.45136	86.38491	rank 9: near Fakirabad (6.7 km), covers 497 more people
SR10	siren site	20.34158	86.38187	rank 10: near Patkura (5.9 km), covers 493 more people
SR11	siren site	20.49497	85.93546	rank 11: Jagatpur, covers 487 more people
SR12	siren site	20.63137	86.22432	rank 12: near Balachandrapur (2.3 km), covers 456 more people
SR13	siren site	20.34478	86.33846	rank 13: near Patkura (8.6 km), covers 392 more people
SR14	siren site	20.40844	86.41295	rank 14: near Patkura (2.2 km), covers 387 more people
SR15	siren site	20.39099	86.44256	rank 15: near Patkura (3.9 km), covers 367 more people
RS01	river sensor bridge	20.45458	85.84814	Devi River bridge, 2823 m, approaches dry in the 25-year flood: yes
RS02	river sensor bridge	20.44052	85.89012	Kathajodi bridge, 869 m, approaches dry in the 25-year flood: yes
RS03	river sensor bridge	20.51335	85.932	Birupa bridge, 543 m, approaches dry in the 25-year flood: yes
RS04	river sensor bridge	20.46498	86.07432	Luna bridge, 776 m, approaches dry in the 25-year flood: no
RS05	river sensor bridge	20.36684	86.26424	Mahanadi bridge, 458 m, approaches dry in the 25-year flood: yes
RS06	river sensor bridge	20.62609	86.32244	Biropa bridge, 424 m, approaches dry in the 25-year flood: no
GB01	bank inspection / soil sa	20.511	85.918	rank 1: behind-bank flood freq 4%, clay loam
GB02	bank inspection / soil sa	20.492	85.832	rank 2: behind-bank flood freq 3%, clay loam
GB03	bank inspection / soil sa	20.429	85.909	rank 3: behind-bank flood freq 6%, clay loam

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
GB04	bank inspection / soil sa	20.429	85.871	rank 4: behind-bank flood freq 7%, clay loam
GB05	bank inspection / soil sa	20.314	86.369	rank 5: behind-bank flood freq 8%, clay loam
GB06	bank inspection / soil sa	20.321	85.881	rank 6: behind-bank flood freq 2%, clay loam
GB07	bank inspection / soil sa	20.294	85.891	rank 7: behind-bank flood freq 2%, clay loam
GB08	bank inspection / soil sa	20.257	85.881	rank 8: behind-bank flood freq 1%, clay loam
GB09	bank inspection / soil sa	20.314	86.398	rank 9: behind-bank flood freq 1%, clay loam
GB10	bank inspection / soil sa	20.314	86.322	rank 10: behind-bank flood freq 7%, clay loam
GB11	bank inspection / soil sa	20.639	86.282	rank 11: behind-bank flood freq 4%, clay loam
GB12	bank inspection / soil sa	20.621	86.33	rank 12: behind-bank flood freq 4%, clay loam
RL01	critical road link	20.26608	86.36142	secondary: 10,930 people's hospital route, 0.83 km flooded (T25)
RL02	critical road link	20.38261	86.1631	unclassified: 10,351 people's hospital route, 1.68 km flooded (T25)
RL03	critical road link	20.27157	86.02512	unclassified: 9,421 people's hospital route, 0.34 km flooded (T25)
RL04	critical road link	20.39945	86.25511	unclassified: 7,657 people's hospital route, 0.14 km flooded (T25)
RL05	critical road link	20.45266	86.37807	tertiary: 7,582 people's hospital route, 0.42 km flooded (T25)
RL06	critical road link	20.34554	86.37918	tertiary: 7,045 people's hospital route, 0.14 km flooded (T25)
RL07	critical road link	20.46508	86.35512	unclassified: 6,210 people's hospital route, 0.34 km flooded (T25)
RL08	critical road link	20.28453	86.44175	residential: 6,171 people's hospital route, 0.03 km flooded (T25)
RL09	critical road link	20.38692	86.42105	tertiary: 4,863 people's hospital route, 0.16 km flooded (T25)
RL10	critical road link	20.52496	86.02748	unclassified: 4,638 people's hospital route, 0.05 km flooded (T25)
WL01	well levels (ground adv	20.3029	86.3751	near Patkura (10.2 km): 37% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL02	well levels (ground adv	20.6188	86.242	Balachandrapur: 30% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL03	well levels (ground adv	20.4029	86.3345	near Patkura (7.6 km): 34% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL04	well levels (ground adv	20.3901	86.4057	Patkura: 75% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL05	well levels (ground adv	20.5662	86.3703	near Nikirai (3.0 km): 46% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL06	well levels (ground adv	20.4113	86.3926	near Patkura (2.7 km): 63% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL07	well levels (ground adv	20.2686	86.3697	near Patkura (14.0 km): 67% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
WL08	well levels (ground adv	20.4333	86.4369	near Patkura (5.8 km): 65% of people within 200 m of 25-yr flood water; measure wells at 0-500 m
HF01	flooded health facility	20.40932	85.93741	PHC, Bentkar
HF02	flooded health facility	20.60406	86.10328	PHC, Samsarpur
HF03	flooded health facility	20.32195	86.42415	PHC, Hansura
HF04	flooded health facility	20.46205	86.35172	Government hospital, Nanakar
HF05	flooded health facility	20.50871	86.41009	Bhumika Hospital

Mahanadi delta field check points (GPX/KML in fieldkit/checkpoints)

id	kind	lat	lon	note
HF06	flooded health facility	20.34104	86.38086	Sub Centre
HF07	flooded health facility	20.51068	85.9146	Unique Hospital